

APOORVA KULKARNI

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SUMMARY

MS CS student at University of Maryland (graduating 2027) pursuing research in multimodal foundation models

EDUCATION

University of Maryland - College Park

2025–2027

Master of Science in Computer Science

(3.93/4.0)

Graduate coursework: Natural Language Processing, LM Interpretability, Systems for ML, Advances in XR

PES University, Bangalore

2020–2024

B.Tech in Computer Science and Engineering

9.69/10 (3.97/4.0)

Awards: Prof. CNR Rao Scholarship (top 2% of cohort in every semester)

SELECTED PUBLICATIONS

[1] **A Closer Look at Failure Modes in Temporal Understanding of Large Audio-Language Models**

Apoorva Kulkarni, Kaousheik J, Sreyan G, Sarah Wiegrefe, Dinesh Manocha, Ramani Duraiswami, **Under Review - ACL 2026**

[2] **FESTA: Functionally Equivalent Sampling for Trust Assessment of Multimodal LLMs**

Debarpan Bhattacharya*, Apoorva Kulkarni*, Sriram Ganapathy, **Findings of EMNLP 2025**

[3] **Benchmarking and Confidence Evaluation of Audio-LLMs For Temporal Reasoning**

Debarpan Bhattacharya*, Apoorva Kulkarni*, Sriram Ganapathy, **INTERSPEECH 2025**

[4] **The Second DISPLACE Challenge: DIarization of SPeaker and LAnguage in Conversational Environments**

Shareef Babu Kalluri, Prachi Singh, Pratik Roy Chowdhuri, Apoorva Kulkarni, et al., **INTERSPEECH 2024**

EXPERIENCE

Research Assistant

Sept 2025 - present

GAMMA Lab - Audio, UMD College Park

- Built a 1,027-question benchmark to diagnose temporal reasoning failures in Large Audio-Language Models, conducted mechanistic attention analysis revealing that models achieve only 58.6-68.75% accuracy due to imprecise attention distribution, and demonstrated layer-specific interventions improve performance

- Leveraging reinforcement learning to enhance temporal alignment and timestamping precision in LALMs

Research Assistant

Jan 2024 – Jun 2025

LEAP Lab, Indian Institute of Science (IISc)

- Proposed FESTA, a black-box unsupervised sampling method for uncertainty estimation in multimodal LLMs via consistency and sensitivity measures. Achieved 30% AUROC improvement on visual and audio reasoning benchmarks

- Co-organized DISPLACE-2024, a special session on multilingual, multispeaker speech tasks at INTERSPEECH 2024. Developed baseline Automatic Speech Recognition models and an evaluation portal for large-scale challenge submissions.

Intern

May 2023 – Aug 2023

Adobe India

- Built and integrated B2B e-commerce features (product request and Q&A modules) for Adobe Commerce, increasing customer engagement and reducing support needed for product queries

SELECTED PROJECTS

Evaluation of Automated Poster Generation

Sept 2025 – Dec 2025

- Created a 15-metric evaluation rubric and 93-poster benchmark to assess VLM-as-a-Judge reliability and model-specific biases for automated poster generation, validated across multiple VLMs and human participants. Results showed that InternVL-3 achieved highest human alignment.

Chain-of-Thought in vision-language models (VLMs)

Jan 2025 – May 2025

- Benchmarked various prompting strategies on spatial reasoning datasets. Designed a two-stage question decomposition pipeline for VLMs, achieving an 8% relative accuracy gain over a Chain-of-Thought baseline.

TECHNICAL SKILLS

Libraries

PyTorch, pandas, numpy, scikit-learn, HuggingFace, NLTK, Keras

Languages

Python, C, Java, JavaScript, HTML/CSS

Machine Learning

pre-training, efficiency, fine-tuning, post-training, reinforcement learning, interpretability